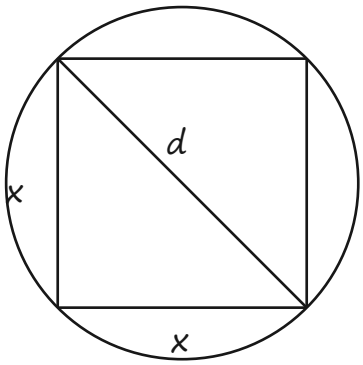
8 A square, with sides of length x cm, is inside a circle.
Each vertex of the square is on the circumference of the circle.

The area of the circle is 49 cm^2.

Work out the value of x.
Give your answer correct to 3 significant figures.



Circle area: $\pi r^2 = 49$

$r^2 = \frac{49}{\pi}$

Diameter $d = 2r$

$d^2 = 4r^2 = \frac{196}{\pi}$

1 mark
Use the circle area to find r or d

The square diagonal is a diameter.

$x^2 + x^2 = d^2$

1 mark
Use Pythagoras

$2x^2 = \frac{196}{\pi}$ so $x^2 = \frac{98}{\pi}$

1 mark
Rearrange for x^2

$x = \sqrt{(98 \div \pi)} = 5.58519...$ cm

1 mark
5.59 cm to 3 significant figures

5.59

(Total for Question 8 is 4 marks)

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